# TIC 973170627

Eclipsing binary candidate dossier - Star Vetter

**THIN MARGIN**

```
BLS EPHEMERIS
  Period       4.514030 d
  Epoch (t0)   1597.0122 BTJD
  Duration     0.1200 d
  Depth        4.473 %  (44,728 ppm)
  SNR          156.5

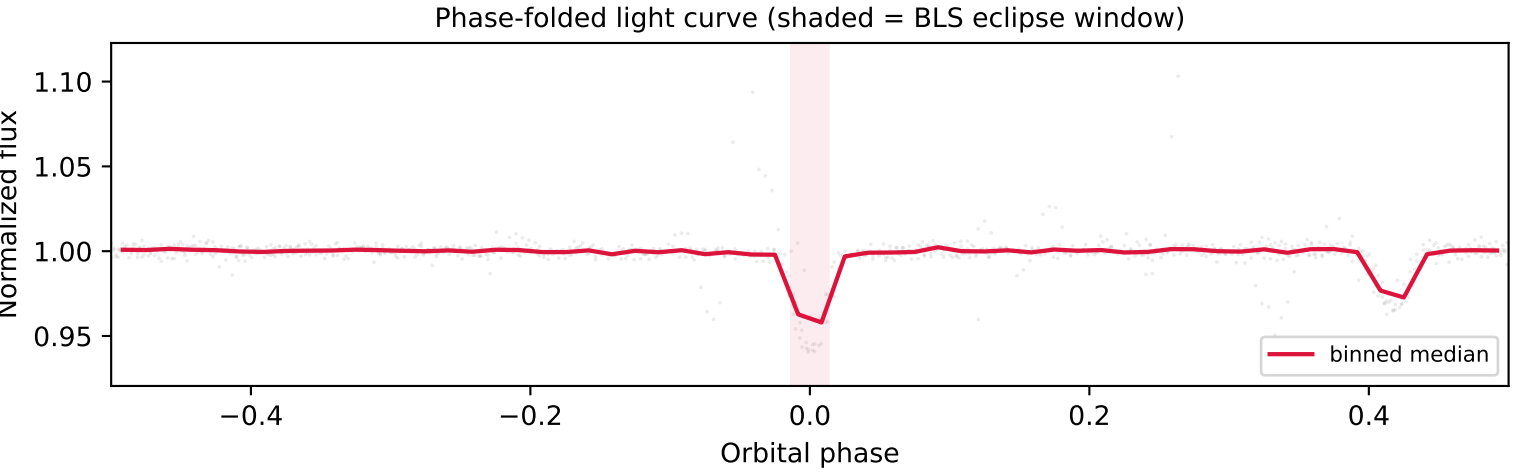
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     5  (gate >= 3)
  Robust depth ratio    1.02  (gate >= 0.50)
  Robust depth sigma    66.4  (gate >= 7.0)
  Odd/even z-score      0.50  (consistent)
  Secondary eclipse     detected, phase 0.41, 6.8 sigma
```

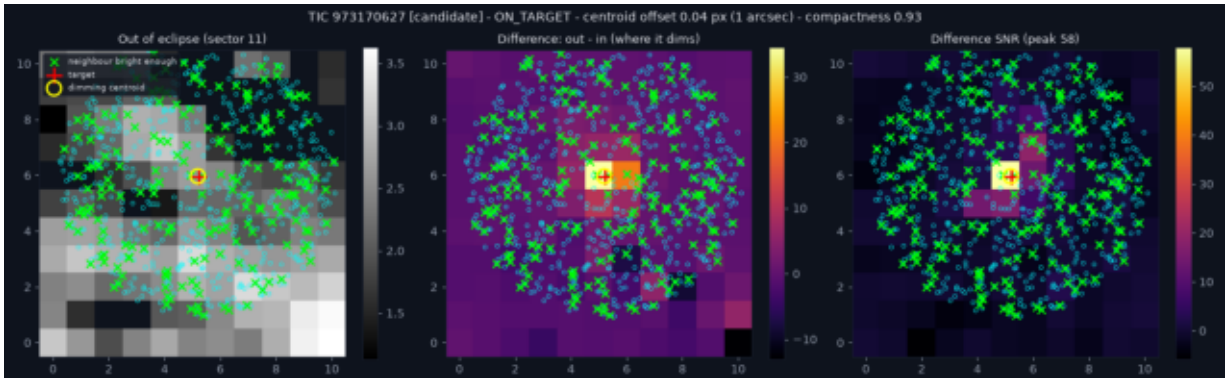
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 11    light-curve dip in this sector: 73.9 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   58.00   (gate >= 5.0)      Compactness   0.93   (gate >= 0.50)
  Dimming centroid offset     0.04 px (1 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 442382112
    0.14 px away, dTmag -1.22 - farther from the dimming than the target
```




Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.097 px (dimming centroid 0.041 px from target vs 0.138 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.